Idea Proposal: Fatura Dedektifi (Bill Detective)

Finding which appliance uses the electricity from a single household meter

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1. Project Idea

An electricity bill shows one total. It does not show which appliance used how much, so a household cannot tell where to save or which tariff suits it. Energy costs are a real burden in Türkiye. In 2025, 19.5% of the population could not afford to keep their home adequately warm [1]. Heating is mostly gas, but electricity is the bill a household can cut appliance by appliance. A separate meter on every appliance is expensive and impractical.

I want to build a model that takes only the total power signal of a home and estimates how much the fridge, the washing machine, the dishwasher, the kettle and the microwave used (non-intrusive load monitoring, NILM). The result will be a web demo that turns one day of meter data into kWh and TL per appliance. It will price the day with Türkiye's single-rate and three-rate (day, peak, night) tariffs, say which one is cheaper, and name the appliance worth moving to the night. I will measure success per appliance with the mean absolute error in watts (MAE) and the error in total energy, on homes the model never saw in training.

2. Relevance to the Sustainable Development Goals

The project serves SDG 7, Affordable and Clean Energy. It relates most to target 7.3 (improve energy efficiency), because people can only reduce the consumption they can see, and more loosely to target 7.1 (affordable energy services), by helping households lower their bills. It also supports SDG 12.8 (information and awareness for sustainable lifestyles) and SDG 13, since lower electricity use means lower emissions.

3. Literature Examples

Kelly and Knottenbelt (2015) [2] applied three deep neural networks to this task for five appliances in UK homes. The networks achieved better average F1 scores than the classical methods (combinatorial optimisation and factorial hidden Markov models) and also worked on a house not seen in training. Zhang et al. (2018) [3] proposed sequence-to-point learning: a convolutional network reads a window of the total signal and predicts the power of one appliance at the midpoint. They report improvements of 84% and 92% on two error measures.

Both papers evaluate models on research benchmarks. Neither turns the result into a tool or a cost breakdown for a household. My project tests on several unseen homes at 1-minute resolution and shows the result in kWh and TL under Turkish tariffs.

4. Data

I will use REFIT Electrical Load Measurements (Cleaned) from the University of Strathclyde, free under CC BY 4.0 [4]. It covers 20 UK homes between 2013 and 2015, with the total power and 9 appliance channels in watts about every 8 seconds, as one CSV file per home (514 MB archive, 6.9 GB and 119.5 million rows unpacked). I downloaded and checked it. A fridge or freezer is monitored in 20 homes, a washing machine in 19, a microwave in 16, a dishwasher in 15 and a kettle in 15. In House 2 the nine monitored appliances explain 32% of the energy and about 25% of the minutes have no data. I will resample to 1 minute, remove windows with gaps, map each home's columns to common appliance names, scale the values, and split by home to avoid leakage.

The homes are British and I found no open appliance-level data from Türkiye, so I will adapt in four steps. Both countries use 230 V at 50 Hz, and the five target appliances are also used in Turkish homes. First, I will test the model on a second dataset. D'Incecco et al. [5] report that a model trained on REFIT has an MAE of 11.8 W on another UK dataset but 32.1 W on US homes, where a model trained locally reaches 23.7 W. Second, I will add loads that are common in Turkish homes and missing in REFIT (air conditioner, electric water heater, tea maker) to the training data artificially. Third, I will evaluate at 1-minute, 15-minute and hourly resolution, because the smart meters now being installed in Türkiye are reported to give hourly data. Fourth, I will measure one Turkish home with smart plugs for a few weeks and report the error on that data.

5. Approach: Deep Learning

I choose deep learning with a machine learning baseline. The input is a raw time series in which appliances overlap and draw similar power: in House 2 the heaters of the washing machine and the dishwasher draw about 2.1 and 2.2 kW. A model has to learn the shape of each appliance's cycle, which suits a 1D convolutional network (sequence-to-point), one small network per appliance, trained on free Colab. I already computed two baselines by training on House 2 and testing on House 5. Predicting the average power gives an MAE of 44.9 W for the kettle and 107.1 W for the dishwasher, and gradient boosting on window statistics lowers these to 17.6 W and 67.6 W. It does not help for the washing machine (44.3 to 43.6 W), because window statistics lose the order of the phases in a cycle, which a convolutional network keeps. My target is to beat gradient boosting on the same test homes for at least four of the five appliances.

For the cost I use EPDK's regulated residential tariff valid from 4 April 2026 [6], with taxes added by calculation. The three-rate tariff costs 3.26 TL per kWh at night (22:00 to 06:00), 4.92 TL in the day and 7.00 TL in the evening peak (17:00 to 22:00). The single-rate tariff costs 3.24 TL up to 8 kWh a day and 4.86 TL above that. I priced one real day of House 2 (10.34 kWh): 37.28 TL on the single rate and 49.95 TL on the three-rate tariff. Moving the dishwasher and washing machine runs to the night lowers the three-rate bill to 45.74 TL, so for this home the single rate stays cheaper. The demo will make the same comparison for any home.

References

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